a

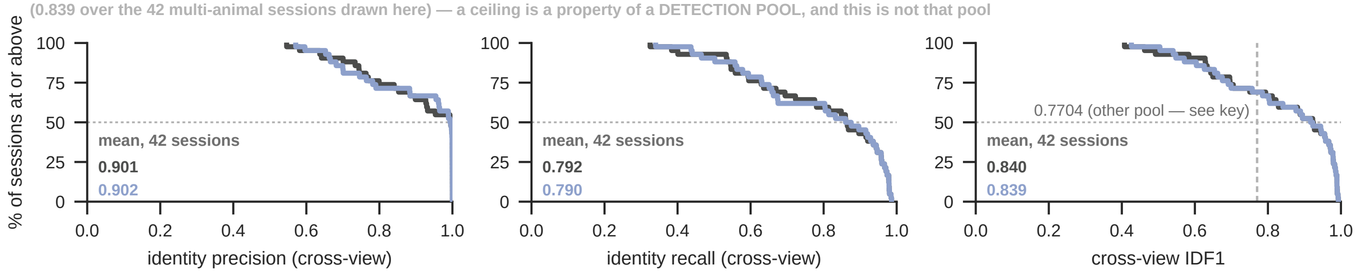
Cross-view identity on 42 multi-animal SLAP-2M sessions

shipped — pose/cross-view-tracker.js as it ships

M1 + stale 20 + distThresh 25 — EXPERIMENTAL: figs/fig8-bench/xv_experimental.js, not in the shipped app

the 42 MULTI-ANIMAL sessions only. The other 32 hold ONE animal, where any tracker scores near-perfectly and both configurations make 0 switches and 0 misgrouped detections, so they are EXCLUDED: pooling them changed only the denominator, a 43% dilution

grey rule 0.7704: the deposit's identity-only ceiling for SLAP-2M, whose detector misses 35.4% of GT (Mouse-Dyad-10M 8.7%, ceiling 0.9527) — a smaller gain here than Mouse-Dyad-10M's 0.749 -> 0.861 is EXPECTED. It is NOT a bound on these curves: that ceiling was measured WITHIN-view on the PAF_3d_kalman pool, where these same 74 sessions score within-view 0.736 (fig3_trackers.json, Fig 7c), against 0.899 on the keeptrack_h5s pool this pass runs on (0.839 over the 42 multi-animal sessions drawn here) — a ceiling is a property of a DETECTION POOL, and this is not that pool



b

Switch and misgrouped-detection rates

shipped — pose/cross-view-tracker.js as it ships

M1 + stale 20 + distThresh 25 — EXPERIMENTAL: figs/fig8-bench/xv_experimental.js, not in the shipped app

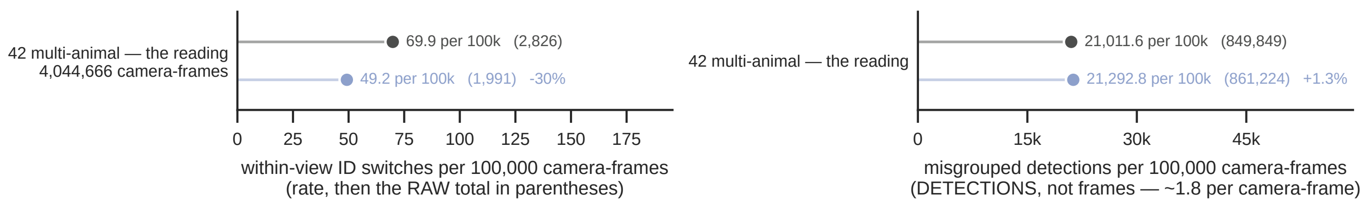
beside each mark: the rate, the RAW total in parentheses, then the change vs shipped. THREE DENOMINATORS, named apart, because two of them have been called “frames”: 674,111 VIDEO FRAMES × 6 cameras = 4,044,666 CAMERA-FRAMES — min(gt, det) per camera, clipped to session length — holding 7,374,061 LABELLED DETECTIONS

READ THE MULTI-ANIMAL COHORT: 32 of the 74 sessions hold ONE animal and contribute 0 switches and 0 misgrouped detections, so they are EXCLUDED — pooling changed only the denominator (43%). misgrouped = a LABELLED DETECTION — one animal, one camera, one frame — over a GT box (IoU ≥ 0.5) whose id disagrees with that box under the OPTIMAL tracker-id -> GT-index permutation for that camera, solved with linear_sum_assignment exactly as IDF1 does internally — 11.5% -> 11.7% of the labelled detections here, a LEVEL readable against IDF1

OTHER SLAP-2M POOL (predictions_h5s, what Fig 7 b-g use), same 42 sessions: misgrouped 303,987 -> 125,142 (-59%), switches 3,094 -> 1,312, cross-view IDF1 0.704 -> 0.721. So the mislabelled-mass result is POOL-DEPENDENT: this arm cuts switches on both pools, mass only there — never mix the pools' levels and the pre-correction claim that this arm made mass WORSE was an artefact of the broken metric, not a finding

corr3dWeight 12, Mouse-Dyad-10M's winner (371 vs 413 switches there), is NEUTRAL there: 1,314 vs 1,312 switches, IDF1 0.7205 vs 0.7212 — corpus-specific

Mouse-Dyad-10M reference (50 sessions, 5 cameras): shipped 2,071 switches (0.00460% of camera-frames) -> stale 20's 413 (0.00092%); cross-view IDF1 0.749 -> 0.861



c

By difficulty and animal count

shipped (pale)

M1 + stale 20 + dt 25 (solid)

42 multi-animal SLAP-2M sessions: 674,111 video frames × 6 cameras = 4,044,666 camera-frames, holding 7,374,061 labelled detections. The 32 one-animal sessions are EXCLUDED: 0 switches and 0 misgrouped detections under both configurations, so pooling them changed only the denominator

misgrouped = a LABELLED DETECTION — one animal, one camera, one frame — over a GT box (IoU ≥ 0.5) whose id disagrees with that box under the OPTIMAL tracker-id -> GT-index permutation for that camera, solved with linear_sum_assignment exactly as IDF1 does internally